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MAAE 2300: Fluid Mechanics I

PA Introduction

Welcome to MAAE 2300 PA tutorials!

☰ Session plan

- 1 **Meet your TA:** background and contact channels.
- 2 **Icebreaker:** quick round to get to know each other.
- 3 **Tutorial approach:** 4-step problem solving framework.
- 4 **Keys to success:** practical tips for assignments and exams.

🤝 My commitment to you

- Tutorials are **interactive workshops**, not passive lectures.
- Safe learning space: questions and mistakes are welcome and expected.
- Focus on developing **physical intuition** rather than rote memorization.

 TA profile

Luis López (please call me Luis)
PhD candidate

› **Research focus:**

- › TSO–DSO coordination and smart grids
- › Optimization and control for energy systems
- › Machine learning for grid flexibility and DERs

› **Technical toolkit:**

- › Mathematical optimization and power system models
- › Julia, Python, and tools for energy analytics
- › Simulation, forecasting, and uncertainty analysis

 Primary channel: email

- › **Official email:**
luislopezdiaz@cmail.carleton.ca
- › **Fastest response:** email is the quickest way to reach me; I typically reply promptly.
- › **Subject tag:** always include [MAAE 2300] for automatic filtering.

 Office hours: by appointment

- › **Flexible scheduling:** consultations are arranged flexibly to fit your schedule.
- › **When needed:** request a meeting for questions requiring in-depth discussion.
- › **Format:** available in person on campus or virtually via MS Teams.

Icebreaker dynamic: breaking the ice!



Round of introductions

Around the room, share with us:

1

Your name

(or preferred name)

2

Your program

(mechanical, aero, civil, etc.)

3

Favorite pick

(movie, TV series, or hobby)



Don't overthink it — whatever show or movie you watched most recently counts!

Problem solving framework

➡ Step 1: Sketch & given data

Draw the system, select control volume (CV) or fluid element, and extract given and target parameters with SI units.

➡ Step 2: Governing laws & assumptions

Select fundamental conservation laws; apply flow assumptions (*steady*, *1D*, *incompressible*) to eliminate inactive terms.

➡ Step 3: Algebra & symbolic derivation

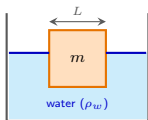
Manipulate equations algebraically to isolate the target variable in symbolic form **before** plugging in numbers!

✓ Step 4: Calculate & sanity check

Substitute numbers with units; perform order-of-magnitude, dimensional, and physical sanity checks.

Framework in action: density & buoyancy

➔ Step 1: Sketch & given data



Given data:

- ▶ Edge: $L = 0.10 \text{ m}$
- ▶ Mass: $m = 0.60 \text{ kg}$
- ▶ Water: $\rho_w = 1000 \text{ kg m}^{-3}$

Target: will it float or sink?

➔ Step 2: Governing laws & assumptions

Definitions: volume $V = L^3$, density $\rho = m/V$, $\text{SG} = \rho/\rho_w$.

Float criterion: floats if $\rho < \rho_w$ ($\text{SG} < 1$), sinks if $\text{SG} > 1$.

Assumption: homogeneous solid with uniform density ρ .

➔ Step 3: Algebra & symbolic derivation

Combine definitions algebraically **before** plugging in numbers:

$$\rho = \frac{m}{L^3}, \quad \text{SG} = \frac{\rho}{\rho_w} = \frac{m}{\rho_w L^3}$$

Analytical condition: the block floats if $\frac{m}{\rho_w L^3} < 1$.

✔ Step 4: Calculate & sanity check

Substitute numbers with consistent SI units:

$$\rho = \frac{m}{L^3} = \frac{0.60 \text{ kg}}{(0.10 \text{ m})^3} = 600 \text{ kg m}^{-3}$$

$$\text{SG} = \frac{\rho}{\rho_w} = \frac{600}{1000} = 0.60$$

Sanity check: $\text{SG} = 0.60 < 1 \implies$ **the block floats** (60% submerged) ✓

Practical keys to acing Fluid Mechanics I

Study and exam habits

- **Understand, do not memorize:** fluid mechanics has fundamental laws; formulas are simply conservation laws applied.
- **Units are your radar:** always track units (N s m^{-2} , Pa, m s^{-1}). If units do not balance, the equation is wrong!
- **Draw first, write later:** a clear, labeled diagram solves half of any fluid mechanics problem.

(Some) classic pitfalls to avoid

- **Gauge vs. absolute pressure:** ideal gas law $P = \rho RT$ and thermodynamics require **absolute pressure** ($P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$).
- **Premature calculation:** rounding intermediate numbers instead of simplifying algebraically first.
- **Skipping assumptions:** applying Bernoulli's equation without verifying irrotational, inviscid, and steady conditions.



Let's have a great term together!